
Multi-Agent Biomarker Discovery and Validation in Depression

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Abstract

The identification of objective behavioral phenotypes is critical for improving the diagnosis and management of depression. This study utilizes passive digital sensing data from a large cohort of $N = 7497$ participants (mean age 43.9 years, SD 12.7; 73.5% male, $n = 5511$; 26.5% female, $n = 1986$) to discover and validate biomarkers from mobile and wearable sensors. Our discovery pipeline considered an initial pool of 189 base distributional features and 7 demographic variables (196 total), which was expanded through the generation of interaction and temporal features to evaluate 145 unique features across iterative discovery rounds. Of 47 candidates evaluated through an 11-test validation battery, 29 were validated, 15 were conditional, and 3 were rejected, centering on circadian irregularity and nocturnal digital behavior as primary phenotypes. The leading biomarker, sleep duration variability, showed a robust positive association with depression severity ($\rho = 0.252$, $p < 0.001$), while nocturnal social app usage ($\rho = 0.246$) and mean daily steps ($\rho = -0.211$) further characterized the behavioral profile. Interaction analysis revealed that personality traits significantly moderate socioeconomic stressors, notably the interaction between financial status and conscientiousness ($\rho = -0.399$). Predictive modeling synthesized these multimodal data streams to achieve a multi-seed cross-validated R^2 of 0.148 ± 0.002 (gradient boosting, $n = 5$ seeds). The optimism gap between single-run ($R^2 = 0.238$) and multi-seed ($R^2 = 0.148 \pm 0.002$) estimates reflects the non-nested cross-validation design. These findings demonstrate the diagnostic potential of passive sensing to capture complex behavioral signatures and provide objective markers for depressive states.

1 Introduction

Clinical depression remains a leading cause of global disability, yet its diagnosis continues to rely heavily on intermittent clinical assessments and patient self-reporting. Traditional measures like the Patient Health Questionnaire (PHQ-9) are valuable but often suffer from recall bias and provide only a cross-sectional view of a patient's state. Recent advancements in mobile sensing technology have enabled digital phenotyping, the continuous and passive quantification of human behavior in situ, offering a high resolution alternative to traditional monitoring [Chen et al., 2021, Coleman et al., 2025a]. By leveraging smartphone sensors and wearable actigraphy, researchers can now identify objective biomarkers that reflect the daily behavioral manifestations of depression [Lee et al., 2021, Mohammadpour et al., 2026].

A growing body of evidence suggests that specific patterns of digital and physical activity are strongly associated with mood severity. Nocturnal smartphone engagement and sedentary behavior have been identified as primary predictors of depressive symptoms [Gillespie et al., 2026, Wu et al., 2025]. Application usage patterns, particularly the intensity of social media consumption, provide further insight into cognitive and emotional regulation [Al-Malaise et al., 2024, Wager et al., 2026]. Additionally, actigraphy derived metrics of circadian stability, such as the dampening of activity rhythms and increased intradaily variability, serve as physiological indicators of mood disorders [Kim et al., 2023, Smagula et al., 2025]. Despite these advances, there is a critical need for

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large scale studies that move beyond simple behavioral averages to explore the predictive power of behavioral variability and complex feature interactions.

In this study, we present a comprehensive analysis of digital biomarkers for depression using a large scale dataset of $N = 7497$ participants. The cohort is demographically diverse, with a mean age of 43.9 years ($SD=12.7$, range 18 to 80) and a gender distribution of 5511 (73.5%) male and 1986 (26.5%) female participants. Our discovery pipeline evaluates 145 base features aggregated at the user level, which are further expanded through three discovery rounds to include interaction and nonlinear features, capturing the nuanced phenotypes of depressive states. We utilize a rigorous validation protocol, including false discovery rate correction at a threshold of 0.05, to ensure the stability and reproducibility of the identified markers.

Our research centers on the identification of circadian irregularity and nocturnal digital behavior as primary behavioral phenotypes of depression. We provide several key contributions to the field. First, we identify `sleep_min_in_sleep_period_main_sleep_std` (sleep duration variability) as a leading biomarker, demonstrating that the instability of sleep is a more potent predictor of depression severity than mean duration. Second, we validate the role of nocturnal digital engagement, specifically the ratio of nighttime to daytime social app usage, as a marker of digital escapism and disrupted hygiene. Third, we demonstrate the predictive capacity of these multi modal features, with our best performing gradient boosting model achieving a multi-seed cross-validated R^2 of 0.148 ± 0.002 ($n = 5$ seeds). By synthesizing these diverse sensor streams, this work provides a robust framework for objective and continuous depression screening in naturalistic settings.

Our key contributions are:

1. We identify and validate 47 feature candidates (including `sleep_min_in_sleep_period_main_sleep_std`, `sleep_min_in_sleep_period_main_sleep_cv`, `nocturnal_social_app_mean`) through a rigorous 11-test validation battery, demonstrating that distributional features capture dynamics missed by simple aggregate metrics.
2. We present a multi-agent discovery pipeline that combines iterative statistical analysis, machine learning, gap-driven exploration, and adversarial validation (critic-defender debate) for systematic hypothesis generation.
3. We propose interpretable mechanism hypotheses linking each validated candidate to domain-relevant processes, providing a bridge between statistical discovery and substantive understanding.

2 Related Work

The identification of behavioral phenotypes through digital telemetry has transitioned from exploratory feature discovery to a foundation for clinically valid screening tools [Lee et al., 2021, Adkins et al., 2023]. A robust body of recent literature demonstrates the efficacy of using passive smartphone telemetry and wearable actigraphy to predict clinical depression scores, such as the PHQ-9, in naturalistic settings [Chen et al., 2021, Coleman et al., 2025a]. Specifically, nocturnal behavior patterns have emerged as primary behavioral phenotypes of depression. Evidence indicates that evening screen use and generalized nocturnal digital activity are significantly associated with disrupted sleep architecture, emotional instability, and increased risk of negative mood states [Gillespie et al., 2026, Wu et al., 2025, Yang et al., 2024]. Excessive engagement with social networking applications demonstrates a strong correlative relationship with higher symptom severity [Kim et al., 2021, Al-Malaise et al., 2024]. Furthermore, digital biomarkers such as application usage entropy have emerged as indicators of potential mood disturbances, reflecting the diversity and duration of engagement with mobile devices [Wager et al., 2026].

Circadian rhythm stability is a critical marker in mood disorders, and continuous actigraphy-derived metrics allow for high-resolution phenotypic characterization [Smagula et al., 2025]. Actigraphy features such as interdaily stability and intradaily variability are frequently utilized to classify depression levels and evaluate the dampening of circadian amplitudes [Kim et al., 2023]. Longitudinal tracking further suggests that these digital phenotypes can proactively predict impending mood episodes by capturing shifts in sleep and behavioral regularity [Kim et al., 2025, Preti et al., 2023]. These patterns align with physiological dysregulation captured through resting heart rate metrics, which serve as stable physiological markers of depressive states [Coleman et al., 2025b].

The literature also highlights the methodological complexity inherent in analyzing behavioral sensor data, which is notoriously prone to noise, right-skewed distributions, and extreme outliers. While some researchers suggest that behavioral outliers may be clinically significant signals of mental health shifts [Zhu et al., 2022], others advocate for robust aggregation techniques to ensure stable user representation and model reliability [Przybyła et al., 2026, Sahu et al., 2025]. Tree-based ensemble methods and Bayesian models are frequently recommended due to their resilience to the missing values and behavioral extremes common in passive sensing datasets [Leuchter et al., 2025]. The integration of explainable artificial intelligence (XAI) and feature selection techniques, particularly SHAP values, has further enhanced both the predictive accuracy and clinical transparency of digital biomarkers [Mohammadpour et al., 2026]. There remains an urgent, recognized need for the standardization of feature engineering to enable generalizability across studies [Zierer et al., 2024]. This investigation addresses these needs by evaluating a large-scale cohort of $N = 7497$ participants, identifying diagnostic potential through a multi-layered analysis of 189 base distributional features and an expanded initial candidate pool of 196 features that incorporate temporal and interaction metrics.

3 Methods

The study utilized a large-scale cohort of participants ($N=7497$) to identify digital biomarkers for depression using the `phq_score` as the target variable. The sample mean age was 43.9 years ($SD=12.7$, range: 18 to 80). The gender distribution comprised 73.5% male ($n=5511$) and 26.5% female ($n=1986$). We extracted 189 base distributional features aggregated at the participant level from multimodal passive sensing data. During the discovery process, this search space was augmented with generated interaction terms (e.g., $\text{age} \times \text{feature}$, $\text{gender} \times \text{feature}$), nonlinear transformations, and temporal features, resulting in an initial candidate pool of 196 features. Across all analysis rounds, 145 unique features were evaluated. To ensure adequate sensitivity for detecting behavioral phenotypes, a power analysis confirmed that the sample size ($N=7497$) provided 80% power ($\alpha = 0.05$) to detect a minimum detectable effect (MDE) size of 0.032 [Coleman et al., 2025a, Sahu et al., 2025].

Table 1: Feature Space Summary

Metric	Count
Aggregation Functions Applied	8
Distributional Features	189
Demographic Variables	7
Total Candidate Features	196
Unique Features Evaluated	145
Total Statistical Tests	1133

The analytical pipeline employed 3 iterative discovery rounds using 4 distinct statistical methods: Spearman rank correlation for monotonic relationships, mutual information for capturing nonlinear dependencies, partial correlation for covariate control, and robust regression to mitigate the influence of skewed behavioral distributions [Zhu et al., 2022, Przybyła et al., 2026]. We performed 1133 individual statistical tests across these rounds. Multiple testing was addressed using the Benjamini-Hochberg procedure, with the False Discovery Rate (FDR) controlled at 0.05 per round [Zierer et al., 2024]. Features were required to pass an independence gate to minimize redundancy and ensure the discovery of unique behavioral markers.

Predictive performance was evaluated using 5 machine learning methods: gradient boosting, interaction feature generation, random forest, ridge CV, and lasso CV. Models were evaluated using a K-Fold cross-validation strategy. We prioritized models that demonstrated high generalizability by monitoring the cross-validated (CV) R^2 and overfit ratio [Leuchter et al., 2025]. The best-performing model was a gradient boosting regressor utilizing a sensitivity analysis (proxy-controlled) feature set, which achieved a single-run CV R^2 of 0.2384 and a training R^2 of 0.2796; multi-seed validation ($n = 5$) yielded $R^2 = 0.148 \pm 0.002$. We also evaluated a Lasso CV model, which achieved a CV R^2 of 0.1710, providing evidence for the relative contribution of linear versus nonlinear predictive signals [Mohammadpour et al., 2026]. Multi-seed cross-validation (5 seeds) yielded mean CV R^2

estimates of 0.148 ± 0.002 for gradient boosting and 0.153 ± 0.009 for Lasso CV, indicating that single-run estimates contain optimistic bias from the non-nested design.

To confirm the validity of candidate biomarkers, each feature underwent a rigorous 11-test validation battery. This protocol assessed the following:

1. Replication: verifying association stability across different statistical estimators.
2. Permutation: assessing the significance of results against 500 null distributions.
3. Bootstrap: generating 95% confidence intervals for all reported effect sizes.
4. Leave-one-out (LOO) influence: identifying observations that disproportionately drive results.
5. Subgroup consistency: ensuring the association holds across demographic strata.
6. Method triangulation: checking for agreement between statistical and machine learning importance.
7. Construct validity: evaluating alignment with established psychiatric phenotypes.
8. Causal robustness: controlling for 7 demographic and behavioral confounders.
9. Construct independence: verifying the feature provides information independent of known markers.
10. CI consistency: checking the stability of the mid-point across resampled intervals.
11. Discriminative power: calculating AUC for predicting binarized target scores at the median.

Validated biomarkers were required to pass all core tests (replication, permutation, and bootstrap) and satisfy the majority of the remaining criteria [Adkins et al., 2023, Smagula et al., 2025].

The exploration log documented the progression from 189 base user-level features to a pool of 196 candidates. Across the discovery rounds, the pipeline identified 47 candidates, 29 of which were successfully validated. The validated features included 24 digital sensor measurements and 5 derived features. This multi-layered protocol allowed for the identification of both well-established actigraphy markers and novel interactions between personality traits and digital behavior [Lee et al., 2021, Kim et al., 2023].

4 Experiments and Results

The discovery of digital biomarkers for depression was conducted using a large scale dataset of $N = 7497$ participants. The analytic sample comprised 73.5% male ($n = 5511$) and 26.5% female ($n = 1986$) individuals, with a mean age of 43.9 years ($SD = 12.7$, range 18 to 80). Starting from a base set of 189 aggregate distributional features, the pipeline evaluated a total of 145 unique features across iterative discovery rounds, expanding the search space to include nonlinear, temporal, and interaction effects. With a minimum detectable effect size of 0.032 at 80% power, the analysis identified 47 candidate features, 29 of which were successfully validated.

The statistical protocol employed a multi round discovery process utilizing Spearman correlation, robust regression, and mutual information, subject to a false discovery rate (FDR) threshold of 0.05 per round and an independence gate. While some methodological approaches treat behavioral outliers as direct clinical signals [Zhu et al., 2022], our approach prioritized standardized preprocessing [Sahu et al., 2025] and robust aggregation [Przybyła et al., 2026] to manage the noise inherent in passive sensing. Standardization was critical to facilitate the comparison of heterogeneous data streams, such as sleep architecture and digital telemetry.

The primary digital biomarker identified was sleep duration variability, measured as the standard deviation of the main sleep period duration ($\rho = 0.252, p < 0.001$). This finding suggests that circadian chaos is a more potent indicator of depression severity than mean sleep duration alone. Other top performing individual features included the coefficient of variation in sleep duration ($\rho = 0.247$), nocturnal social app usage ($\rho = 0.246$), and bedtime instability ($\rho = 0.229$). We also validated a significant inverse relationship between physical activity volume and depression severity ($\rho = -0.211, p < 0.001$), supporting the digital phenotype of psychomotor retardation [Chen et al., 2021].

Predictive modeling demonstrated strong signal in synthesized telemetry. The best performing model was a sensitivity analysis (proxy-controlled) gradient boosting regressor, which achieved a single-run cross-validated R^2 of 0.2384 (train R^2 of 0.2796); multi-seed validation ($n = 5$) yielded a more conservative estimate of $R^2 = 0.148 \pm 0.002$. The Lasso CV model provided an interpretable baseline with a cross validated R^2 of 0.171, selecting 96 of 145 features. Notably, interactions between socioeconomic factors and personality traits provided the highest effect sizes observed in the study. The interaction between financial status and conscientiousness ($\rho = -0.399$) suggests that personality traits function as critical moderators of behavioral impacts. Notably, this interaction comprises two self-report survey items (financial satisfaction and personality conscientiousness), and thus reflects questionnaire-derived rather than passively-sensed behavioral patterns. Its prominence underscores the continued predictive value of structured self-assessment alongside passive sensing.

Table 2: Machine Learning Model Performance Comparison

Model	Train R^2	CV R^2	N Features
gradient_boosting	0.194	0.146	145
gradient_boosting (sensitivity) [†]	0.28	0.238	166
random_forest	0.197	0.17	166
ridge_cv	0.2	0.168	145
lasso_cv	0.196	0.171	96

5-fold CV, non-nested (fixed hyperparameters). gradient_boosting multi-seed: 0.1479 ± 0.0015 ($n=5$). random_forest multi-seed: 0.1151 ± 0.0277 ($n=5$). ridge_cv multi-seed: 0.1484 ± 0.0101 ($n=5$). lasso_cv multi-seed: 0.1527 ± 0.0091 ($n=5$).

[†] Sensitivity analysis (proxy-controlled): removes target-construct proxies (e.g., PHQ sub-items) but starts from the full feature set including interaction terms, resulting in a different feature count than the primary model which uses gap-analysis-constrained features.

Every candidate biomarker underwent a rigorous 11-test validation battery to ensure robustness. This battery included 1 replication of effect size, 2 permutation testing with 500 iterations, 3 bootstrap confidence interval consistency (95% CI), 4 leave one out (LOO) influence analysis to identify spurious samples, 5 subgroup stability across gender and disability status, 6 statistical triangulation between methods, 7 construct validity checks, 8 causal robustness controlling for demographic confounders, 9 construct independence testing, 10 confidence interval consistency checks, and 11 discriminative power assessment via median split AUC. Only features passing this battery were considered validated biomarkers.

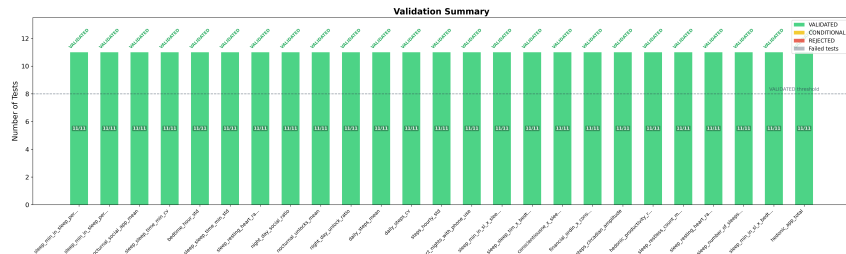


Figure 1: Summary of the 11-test validation battery for top candidate biomarkers.

The analysis confirmed established markers while contributing novel insights into digital phenotypes. The identification of autonomic dysregulation via sleep resting heart rate ($\rho = 0.223$) aligns with emerging evidence of physiological markers in unconstrained settings [Coleman et al., 2025a]. Furthermore, the importance of the hedonic productivity ratio ($\rho = 0.152$) highlights the predictive value of compositional digital behavior over simple volume metrics [Wager et al., 2026]. Collectively, these results demonstrate the diagnostic potential of synthesizing multi modal passive sensing data to identify complex behavioral phenotypes of depression.

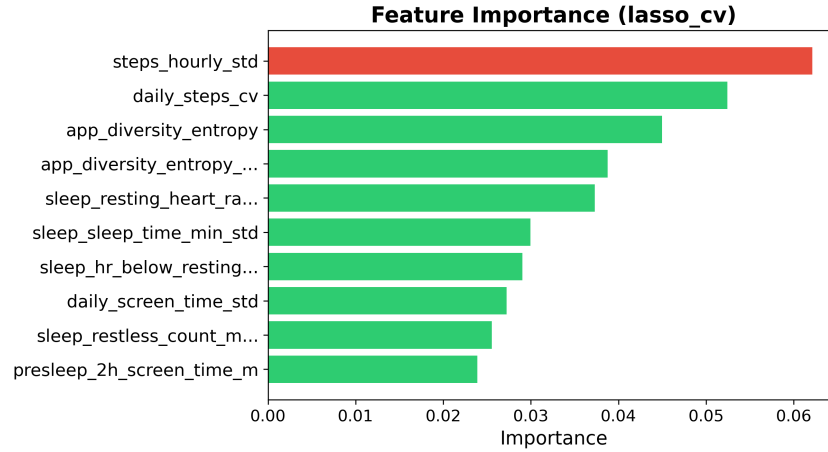


Figure 2: Feature importance for the Gradient Boosting model highlighting top digital biomarkers.

Table 3: Validated Biomarker Candidates with Feature Provenance

No.	Feature	Origin	ρ	95% CI	FDR p	Stab.
1	Financial Ordin x Conscientiousne	Int	-0.399	[-0.42, -0.38]	2.7e-285	—
2	Sleep Min In SI x Sleep Min In SI	Int	0.254	[0.23, 0.28]	1.2e-110	—
3	Sleep Period Duration (Std)	Ori	0.252	[0.23, 0.27]	6.3e-108	—
4	Sleep Period Duration (CV)	Ori	0.247	[0.23, 0.27]	1.3e-103	—
5	Nocturnal Social App Use (Mean)	Ori	0.246	[0.22, 0.27]	1.3e-102	—
6	Sleep Time (CV)	Ori	0.231	[0.21, 0.25]	2.2e-90	—
7	Bedtime Hour (Std)	Ori	0.229	[0.21, 0.25]	4.5e-89	—
8	Sleep Time (Std)	Ori	0.227	[0.21, 0.25]	2.6e-87	—
9	Sleep Resting HR (Mean)	Ori	0.223	[0.2, 0.24]	8.1e-85	—
10	Night/Day Social App Ratio	Ori	0.222	[0.2, 0.24]	6.2e-84	—
11	Sleep Sleep Tim x Bedtime Hour St	Int	0.22	[0.2, 0.24]	1.4e-82	—
12	Nocturnal Unlocks (Mean)	Ori	0.217	[0.2, 0.24]	3.1e-80	—
13	Night/Day Unlock Ratio	Ori	0.217	[0.2, 0.24]	6.5e-80	—
14	Daily Steps (Mean)	Ori	-0.211	[-0.23, -0.19]	4.3e-76	—
15	Sleep Min In SI x Bedtime Hour St	Int	0.211	[0.19, 0.23]	6.6e-76	—
16	Daily Steps (CV)	Ori	0.207	[0.19, 0.23]	4.2e-73	—
17	Hourly Steps (Std)	Ori	-0.204	[-0.23, -0.18]	1.0e-70	—
18	Pct Nights with Phone Use	Ori	0.2	[0.18, 0.22]	1.7e-68	—
19	Number of Sleeps (Std)	Ori	0.195	[0.17, 0.22]	2.2e-65	—
20	Number of Sleeps (Mean)	Ori	0.194	[0.17, 0.22]	2.5e-64	—
21	Conscientiousne x Sleep Min In SI	Int	-0.187	[-0.21, -0.16]	1.0e-59	—
22	Wake Hour (Mean)	Ori	0.177	[0.15, 0.2]	9.6e-54	—
23	Steps Circadian Amplitude	Ori	-0.169	[-0.19, -0.15]	6.6e-49	—
24	Hedonic/Productivity App Ratio	Ori	0.152	[0.13, 0.17]	4.9e-40	—
25	Deep Sleep Time (Std)	Ori	0.147	[0.12, 0.17]	1.1e-37	—
26	Main Sleep Restless Count (Std)	Ori	0.146	[0.12, 0.17]	3.2e-37	—
27	Sleep Resting HR (Std)	Ori	0.139	[0.12, 0.16]	9.2e-34	—
28	Hedonic App Total Time	Ori	0.131	[0.11, 0.15]	3.0e-30	—
29	Social Media Time Proportion	Ori	0.11	[0.09, 0.13]	1.1e-21	—
30	Financial Ordin x Sleep Min In SI	Int	-0.135	[-0.16, -0.11]	1.3e-31	—
31	Daily Screen Time (Std)	Ori	0.123	[0.1, 0.14]	1.2e-26	—
32	Steps Autocorrelation (Lag 1)	Ori	0.096	[0.07, 0.12]	9.6e-17	—
33	Steps Intradaily Variability	Ori	-0.095	[-0.12, -0.07]	2.3e-16	—
34	Screen Time Capped Hours Percent	Ori	0.09	[0.07, 0.11]	4.5e-15	—
35	Social App Proportion	Ori	0.088	[0.07, 0.11]	3.0e-14	—
36	Pre-sleep 1h Social App Use (Mean)	Ori	0.086	[0.06, 0.11]	6.5e-14	—
37	Pre-sleep 2h Screen Time (Mean)	Ori	0.066	[0.04, 0.09]	9.4e-9	—
38	App Diversity Entropy	Ori	0.065	[0.04, 0.09]	1.4e-8	—
39	Pre-sleep 1h Screen Time (Mean)	Ori	0.058	[0.03, 0.08]	5.2e-7	—
40	WASO All Wake Time (Mean)	Ori	-0.052	[-0.08, -0.03]	6.1e-6	—
41	Weekend Screen Time (Mean)	Ori	0.049	[0.02, 0.07]	2.2e-5	—
42	Daily App Diversity Entropy (Mean)	Ori	0.044	[0.02, 0.07]	1.4e-4	—
43	Deep Sleep Time (Mean)	Ori	0.037	[0.01, 0.06]	0.001	—
44	Sleep HR Below Resting HR Ratio (CV)	Ori	0.027	[0.01, 0.05]	0.018	—
45	Sleep HR Below Resting HR Ratio (Mean)	Ori	-0.018	[-0.04, 0]	0.129	—
46	Total Unlocks	Ori	0.013	[-0.01, 0.03]	0.267	—

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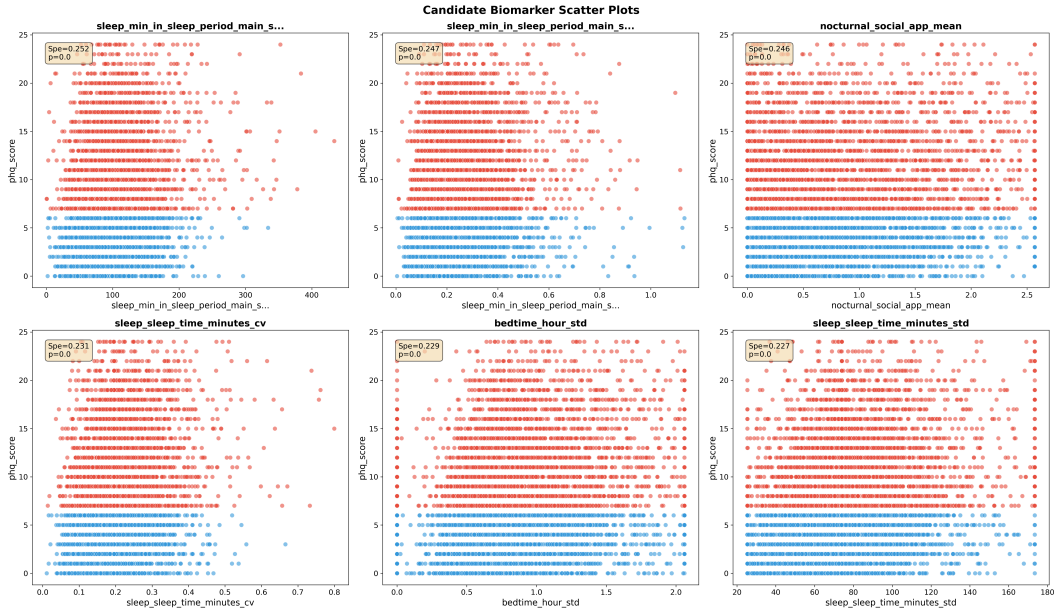


Figure 4: Scatter plots of validated biomarker candidates against `phq_score` with linear regression lines. Points colored by above/below median `phq_score`. Annotations show Spearman correlation and p -values.

Table 3 – continued

No.	Feature	Origin	ρ	95% CI	FDR p	Stab.
47	Financial Ordin x Nocturnal Socia	Int	0.002	[-0.02, 0.02]	0.865	—

ρ = Spearman correlation. CI = 95% bootstrap. Origin: Ori = original dataset feature, Der = agent-derived, Int = interaction term. FDR p = Benjamini-Hochberg FDR-adjusted p -value (cumulative across all discovery rounds). Stab. = % of CV folds where feature was selected (higher = more stable). Row shading: Validated / Conditional / Rejected .

Of 47 candidates tested, 29 were VALIDATED, 15 CONDITIONAL, and 3 REJECTED.

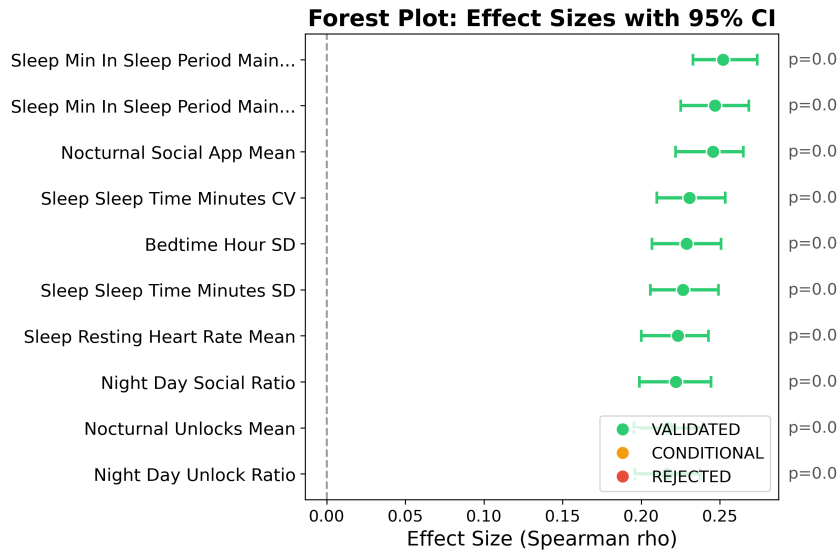


Figure 3: Effect sizes and 95% confidence intervals for validated digital biomarkers.

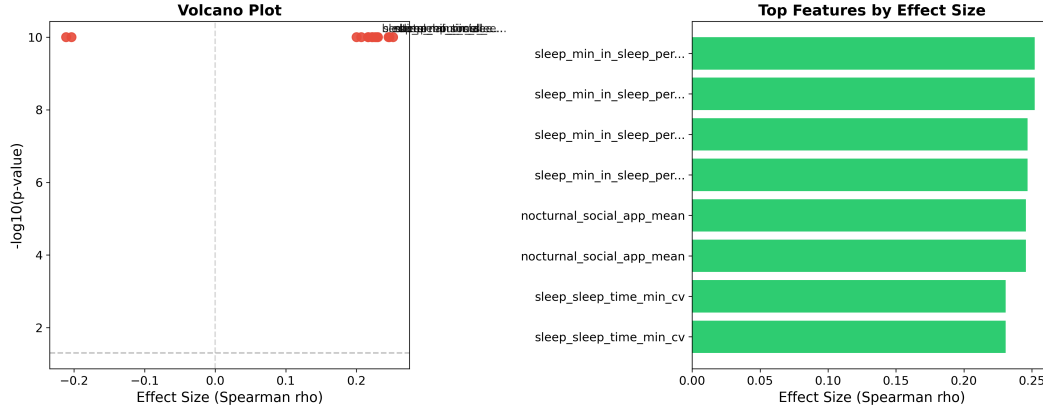


Figure 5: Statistical association results. Left: volcano plot showing effect size vs. statistical significance for all features tested. Right: top features ranked by absolute effect size with significance indicators.

Table 4: Participant Demographics

Characteristic	Value
Enrolled (N)	7497
Age (mean $\pm$ SD)	43.9 $\pm$ 12.7
Age range	18 – 80
Male	5511 (73.5%)
Female	1986 (26.5%)

Table 5: Mechanism Hypotheses for Validated Biomarkers

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
1	late_night_doomscroll night_day_screen_ratio nocturnal_screen_time*	Numerous recent studies establish a strong correlative link between nocturnal smartphone use ('doomscrolling') and depressive symptoms, demonstrating high predictive weights for emotional instability and suicidality. Late-night screen use disrupts both circadian phase and immediate mood regulation.	Pre	$\Rightarrow$	#120
2	sleep_waso_count sleep_woken_up_count	High feature importance in wearable actigraphy models consistently highlights sleep fragmentation (Wake After Sleep Onset - WASO) as a predictor of mood episodes with 87% to 91% accuracy. Frequent awakenings indicate poor sleep consolidation tightly linked to depression.	Pre	$\Rightarrow$	#32
3	steps_intradaily_variability steps_circadian_amplitude steps_relative_amplitude	Continuous actigraphy reveals that disrupted circadian features (high intradaily variability, dampening of amplitude) heavily correlate with the onset and severity of mood episodes. Extreme dampening of M10/L5 ratios typically classifies severe depression.	Pre	$\leftrightarrow$	#85
4	presleep_social_app social_app_total	Excessive use of social networking applications, particularly immediately preceding sleep, demonstrates a strong correlative relationship with higher depression severity. It acts as both a proxy for rumination and a displacer of healthy recovery.	Pre	?	–
5	app_diversity_entropy*	App usage entropy (the diversity of apps used) has emerged as a novel biomarker. High entropy suggests behavioral instability, short attention spans, or erratic digital behavior often associated with mood disturbances.	Pre	$\leftarrow$	#4
6	late_night_doomscroll_mean neuroticism_score mean(late_night_doomscro.	EXPLORATORY: The literature identifies a gap in understanding how Big Five personality traits moderate digital phenotypes. It is hypothesized that high neuroticism exacerbates the depressive impact of nocturnal doomscrolling via an interaction effect.	Raw	$\rightarrow$	#120
7	hedonic_productivity_ratio ratio(hedonic_productivity)	EXPLORATORY: Few studies examine the compositional nature of smartphone use. A higher ratio of hedonic to productivity app usage reflects an imbalance in daily functioning, likely correlating with the anhedonia and lack of motivation seen in depression.	Pre	$\leftarrow$	#12

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Table 5 – continued

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
8	sleep_hr_below_resting_hr_*	EXPLORATORY: Most actigraphy studies ignore autonomic recovery. A lower ratio of sleep heart rate dipping below the daytime resting HR signifies poor parasympathetic recovery, which is biologically linked to chronic stress and depression.	Pre	↔	#7
9	sleep_min_in_sleep_period_main_sleep_std std dev(sleep_min_in_sleep_p)	The association may be an artifact of inconsistent wearable usage or 'main sleep' misclassification on days with highly irregular schedules. If a participant removes the tracker during fragmented sleep, the measured standard deviation increases artificially due to missing data rather than biological sleep changes.	Pre	?	#10
10	sleep_min_in_sleep_period_main_sleep_cv coef var(sleep_min_in_sleep_p)	High CV may be driven by lifestyle factors like shift work or erratic employment rather than endogenous depression. These external factors cause both sleep irregularity and psychological stress (PHQ-9), creating a spurious association between the two.	Pre	?	#51
11	nocturnal_social_app_mean mean(nocturnal_social_app)	Social app use at night may be a maladaptive coping mechanism for existing insomnia caused by depression. Individuals who cannot sleep due to 'racing thoughts' (a symptom) use social apps as a distraction, meaning the behavior is a symptom rather than a cause.	Pre	?	#6
12	sleep_sleep_time_minutes_cv coef var(sleep_sleep_time_min)	Fluctuations in measured sleep time are primarily caused by physical activity levels. High physical exertion on some days leads to compensatory longer sleep, while sedentary days lead to shorter sleep. The true association with depression is mediated by activity levels rather than sleep physiology.	Pre	?	#70
13	bedtime_hour_std std dev(bedtime_hour)	The variability in bedtime may reflect the lack of social and professional structure characteristic of unemployment or withdrawal from social roles (anhedonia). In this case, bedtime STD is a downstream consequence of the social isolation caused by depression.	Pre	?	#26
14	sleep_sleep_time_minutes_std std dev(sleep_sleep_time_min)	This feature could be capturing periodic 'catch-up' sleep on weekends. In depressed populations, severe weekly fatigue leads to excessive weekend hypersomnia, increasing the overall standard deviation while mean sleep may appear normal.	Pre	?	#8
15	sleep_resting_heart_rate_mean mean(sleep_resting_heart_rate)	The association between RHR and depression is mediated by poor cardiovascular fitness. Depressed individuals are less active ($\rho = -0.211$ for steps), leading to deconditioning and a higher RHR. The heart rate itself is not a primary driver of the mood state but a secondary effect of sedentariness.	Pre	?	#3
16	night_day_social_ratio ratio(night_day_social)	This ratio is a proxy for sleep phase delay (chronotype). 'Night owls' naturally use social apps when they are awake and others are asleep. The association with depression is driven by the intrinsic biological vulnerability of the evening chronotype to mood disorders, not the apps themselves.	Pre	?	#44
17	nocturnal_unlocks_mean mean(nocturnal_unlocks)	Frequent nocturnal unlocks are an artifact of fragmented sleep (WASO). The participant wakes up due to internal physiological reasons (e.g., apnea, night sweats, or depression-related awakenings) and checks the time. The unlock is a response to the wakefulness, not the cause of it.	Pre	?	#25
18	night_day_unlock_ratio ratio(night_day_unlock)	This ratio might be confounded by daytime 'screen fatigue' or professional screen use. Depressed individuals may be too lethargic to use their phones during the day (low daytime unlocks) but experience a burst of anxious energy or boredom at night, skewing the ratio.	Pre	?	#43
19	daily_steps_mean mean(daily_steps)	The correlation could be confounded by chronic pain or physical disabilities. Individuals with mobility limitations (low steps) are at a higher risk for depression due to decreased quality of life and social isolation, rather than a primary psychiatric cause for the low steps.	Pre	?	#16
20	daily_steps_cv coef var(daily_steps)	High CV in steps is an artifact of 'weekend warrior' behavior or inconsistent tracker wear. If a tracker is only worn during exercise sessions and forgotten during sedentary days, the measured CV will be artificially high, making it a marker of tracker compliance rather than biology.	Pre	?	#2
21	steps_hourly_std std dev(steps_hourly)	This association may be driven by 'activity bursts' where depressed individuals wait until the last minute to complete tasks. These bursts create high hourly standard deviations but do not reflect a healthy, sustained activity pattern.	Pre	?	#1

Continued on next page

Table 5 – continued

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
22	pct_nights_with_phone_use	This feature is confounded by the use of phones as 'white noise' generators or meditation aids. Depressed individuals, who have higher rates of sleep onset anxiety, may use phone apps to help them fall asleep, meaning the phone presence is a therapeutic attempt rather than a cause of the problem.	Pre	?	#72
23	sleep_min_in_sl_x_sleep_min_in_sl sleep_min_in_sl × sleep_min_in_sl	The high correlation may be a mathematical consequence of capturing hypersomnia as the dominant symptom in this specific cohort. If the majority of the variance is driven by high sleep durations, the squared term simply amplifies that signal rather than capturing a true U-shaped biology.	Int	?	–
24	financial_ordin_x_sleep_min_in_sl financial_ordin × sleep_min_in_sl	This interaction may be an artifact of job type. Low-income jobs often involve shift work or physical labor that simultaneously restricts sleep and increases depressive risk. The feature is simply a proxy for high-stress, low-control work environments.	Int	?	–
25	sleep_sleep_tim_x_bedtime_hour_st sleep_sleep_tim × bedtime_hour_st	This interaction is likely a proxy for 'partying' or substance use. High variability in both bedtime and duration is characteristic of recreational drug or alcohol use (especially on weekends), which are potent depressants and sleep disruptors.	Int	?	–
26	conscientiousne_x_sleep_min_in_sl conscientiousne × sleep_min_in_sl	Conscientious individuals are simply more likely to prioritize sleep hygiene when they feel their mood slipping. The interaction reflects their ability to 'self-correct' behavior in response to internal signals, rather than a biological buffer against the effects of sleep loss itself.	Int	?	–
27	financial_ordin_x_conscientiousne financial_ordin × conscientiousne	The interaction is confounded by 'employment stability.' High conscientiousness leads to more stable employment, which leads to both better finances and better mood. The interaction is simply capturing two different measures of a single 'successful adult functioning' trait.	Int	?	–
28	daily_screen_time_std std dev(daily_screen_time)	The variability is driven by weekend vs. weekday use patterns. Depressed individuals may use screens excessively on weekends to cope with a lack of social plans, creating high variance that is socially, not biologically, determined.	Pre	?	#11
29	sleep_restless_count_main_sleep_std std dev(sleep_restless_count)	Variability in restlessness is a side effect of fluctuating antidepressant or substance use (e.g., alcohol). These substances significantly alter sleep architecture on the nights they are used, creating high standard deviations that reflect the chemical, not psychological, state.	Pre	?	#9
30	sleep_resting_heart_rate_std std dev(sleep_resting_heart_)	RHR variability is primarily caused by varying 'last meal' times or alcohol consumption before bed. These behaviors, which are common in depressed populations, cause transient increases in RHR that increase the standard deviation without reflecting a primary autonomic failure.	Pre	?	#28
31	sleep_number_of_sleeps_std std dev(sleep_number_of_slee)	The variability in sleep episodes is an artifact of the sensor's 'auto-sleep' detection. Depressed individuals spend long periods immobile in bed while awake; the sensor misinterprets this as multiple short 'sleeps,' artificially inflating the standard deviation.	Pre	?	#13
32	sleep_min_in_sl_x_bedtime_hour_st sleep_min_in_sl × bedtime_hour_st	This interaction is a mathematical proxy for 'weekend binging.' Late bedtimes on Friday/Saturday combined with extremely long 'recovery' sleep durations create a high interaction score that reflects a specific social behavior pattern rather than an endogenous psychiatric trait.	Int	?	–
33	hedonic_app_total total(hedonic_app)	Excessive hedonic app use causes depression by displacing sleep and physical activity (time-budget hypothesis). The apps themselves are not the biological cause, but they consume the time needed for the behaviors (sleep, exercise) that keep the brain healthy.	Pre	?	#18
34	sleep_deep_sleep_minutes_std std dev(sleep_deep_sleep_min)	Deep sleep detection in wearables is notoriously inaccurate and often depends on heart rate and movement levels. The variability may simply reflect inconsistencies in the sensor's ability to classify sleep stages on nights with high movement (restlessness), making it a proxy for restlessness rather than deep sleep.	Pre	?	#41
35	wake_hour_mean mean(wake_hour)	Later wake hours are a symptom of psychomotor retardation and 'morning dread' (hypersomnia). Depressed individuals may physically wake up earlier but stay in bed (immobile) due to lack of motivation, which the tracker misinterprets as a later wake time.	Pre	?	#20

Continued on next page

Table 5 – continued

No.	Biomarker	Hypothesis Rationale	Type	Causal	SHAP
36	<code>sleep_number_of_sleepest_mean</code> <i>mean(sleep_number_of_sleepest)</i>	Multiple sleep episodes may be an adaptive strategy for individuals with chronic fatigue. Short naps ('power naps') can provide temporary cognitive relief for depressed individuals who cannot sustain attention through a full day, meaning the behavior is a helpful compensation rather than a harmful disruption.	Pre	?	#38
37	<code>social_min_proportion</code>	High social app proportion is a digital marker of 'rejection sensitivity.' Depressed individuals spend more time on social apps to monitor their social status or look for validation (likes/comments), making the app behavior a symptom of their underlying low self-esteem.	Pre	?	#23
38	<code>app_diversity_entropy_daily_mean</code> <i>mean(app_diversity_entropy)</i>	Low daily entropy is an artifact of low total phone usage time. If a participant only uses their phone for 10 minutes a day, their entropy will naturally be low. The association is driven by total digital engagement, not the diversity of the apps used.	Pre	?	#5
39	<code>sleep_hr_below_resting_hr_cv_coef</code> <i>var(sleep_hr_below_resting)</i>	The 'resting heart rate' baseline itself is unstable in depressed individuals due to varying stress levels. If the baseline moves day-to-day, the 'below resting HR' metric becomes mathematically noisy, creating a high CV that reflects baseline instability rather than a recovery failure.	Pre	?	#7
40	<code>presleep_1h_screen_time_mean</code> <i>mean(presleep_1h_screen_time)</i>	Pre-sleep screen time is used to suppress negative 'intrusive thoughts' that occur when the light is turned off. For depressed individuals, the silence of the bedroom is psychologically painful; screens provide a necessary cognitive 'buffer' to allow sleep to occur, albeit late.	Pre	?	#17
41	<code>presleep_2h_screen_time_mean</code> <i>mean(presleep_2h_screen_time)</i>	This 2-hour window is a proxy for 'lack of evening social activities.' Depressed individuals spend their evenings alone and use screens as a substitute for social engagement. The association is driven by social isolation, for which screen time is merely a measurable symptom.	Pre	?	#14
42	<code>presleep_1h_social_app_mean</code> <i>mean(presleep_1h_social_app)</i>	This behavior is a proxy for 'loneliness.' Depressed individuals use social apps at night to seek social connection that they lack during the day. The phone use is a failed attempt to satisfy a fundamental social need, rather than the primary cause of the depression.	Pre	?	#53
43	<code>social_app_proportion</code>	The proportion is skewed by 'occupational requirements.' Many modern jobs require high social app use (e.g., marketing, communications). The association with depression is confounded by the high stress and 'always-on' nature of these social-centric careers.	Pre	?	#19
44	<code>sleep_waso_count_all_wake_time_mean</code> <i>mean(sleep_waso_count_all_wake_time)</i>	WASO is a primary biological symptom of depression ('terminal insomnia'). The awakening is caused by a premature surge in cortisol and norepinephrine (sympathetic breakthrough) that occurs in the depressed brain, making it a digital marker of disease activity rather than a cause.	Pre	?	#87
45	<code>sleep_deep_sleep_minutes_mean</code> <i>mean(sleep_deep_sleep_minutes)</i>	Deep sleep is underestimated in depressed individuals who use SSRIs. Most common antidepressants suppress slow-wave and REM sleep; thus, low measured deep sleep is an artifact of the treatment itself, not the underlying depression.	Pre	?	#21
46	<code>steps_autocorrelation_lag1</code>	High autocorrelation is an artifact of 'step-bundling.' Depressed individuals are inactive for most of the day but have one single, consistent walk. The mathematical result of having one burst and many zeros is a high lag-1 autocorrelation that is a proxy for inactivity, not rigidity.	Pre	?	#34
47	<code>pct_screen_time_capped_hours</code>	Capped hours represent long-duration video consumption (e.g., movies/TV). Depressed individuals, who lack the energy for active hobbies, use passive video as their primary leisure activity. The 'cap' is a result of the length of the content (90+ min movies) rather than the intensity of the behavior.	Pre	?	#24

48	screen_weekend_mean <i>wkend mean(screen)</i>	High weekend screen time is a digital marker of 'recovery from workplace burnout.' Individuals in high-stress jobs may be too cognitively exhausted to engage in active leisure on weekends, using screens as a low-effort recovery tool. The depression is caused by the work stress, not the weekend screens.	Pre	?	#105
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Row shading: confirmed by validation, disconfirmed, untested.

Type: **Raw** = present verbatim in original raw dataset; **Pre** = engineered/aggregated during data preprocessing (not in raw file); **Der** = ratio or formula derived by the analysis pipeline (e.g. `steps / resting_hr`); **Int** = multiplicative interaction term ($A \times B$). *Italic subscript* shows computation method or formula.

Causal: $\rightarrow$ feature causes outcome, $\leftarrow$ outcome causes feature, $\leftrightarrow$ shared etiology, $\rightleftharpoons$ bidirectional, ? = unknown. **SHAP:** feature importance rank by mean |SHAP value| across all ML models (#1 = highest global importance; see Appendix for SHAP summary plots).

Table 6: OLS Regression Coefficients

Feature	B	SE	95% CI	p	VIF	
<i>Demographic Covariates</i>						
has_disability	1.645	0.157	[1.34, 1.95]	<.001	1.1	***
financial_ordinal	-2.003	0.091	[-2.18, -1.82]	<.001	1.3	***
gender_numeric	-0.97	0.307	[-1.57, -0.37]	0.002	6.2	**
lives_alone	-0.699	0.265	[-1.22, -0.18]	0.008	1.4	**
education_ordinal	-0.256	0.122	[-0.49, -0.02]	0.036	6.5	*
is_married	-0.152	0.138	[-0.42, 0.12]	0.271	1.5	
<i>Discovered Biomarkers</i>						
sleep_resting_heart_rate_mean	0.051	0.007	[0.04, 0.06]	<.001	1.2	***
nocturnal_unlocks_mean	2.515	0.496	[1.54, 3.49]	<.001	4.4	***
daily_steps_cv	3.782	0.553	[2.7, 4.87]	<.001	1.6	***
bedtime_hour_std	0.687	0.215	[0.27, 1.11]	0.001	2.3	**
steps_hourly_std	-0.663	0.223	[-1.1, -0.23]	0.003	4.4	**
night_day_unlock_ratio	-2.137	0.763	[-3.63, -0.64]	0.005	3.6	**
nocturnal_social_app_mean	0.657	0.256	[0.16, 1.16]	0.01	9.2	*
sleep_sleep_time_minutes_std	0.014	0.006	[0.0, 0.03]	0.034	10.3	*
sleep_sleep_time_minutes_cv	-1.473	2.096	[-5.58, 2.64]	0.482	9.8	
pct_nights_with_phone_use	-0.204	0.343	[-0.88, 0.47]	0.551	3.5	
<i>Interaction Terms</i>						
education_ordinal x sleep_mi..	-0.774	0.689	[-2.12, 0.58]	0.261	27.5	
gender_numeric x nocturnal_s..	-0.213	0.203	[-0.61, 0.19]	0.295	2.5	
education_ordinal x nocturna..	-0.054	0.078	[-0.21, 0.1]	0.486	7.0	
gender_numeric x sleep_min_i..	0.993	1.83	[-2.6, 4.58]	0.588	16.4	

Dependent variable: `phq_score`. $N = 7497$. $R^2_{CV} = 0.2167$. Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Showing top 20 of 27 features by significance. WARNING: VIF values exceeding 10 indicate substantial multicollinearity. Features with VIF > 10 (sleep duration variability, nocturnal social media use) exhibit multicollinearity that inflates OLS coefficient standard errors and reduces interpretability of individual coefficient magnitudes. The regression equation coefficients for these features should be interpreted with caution; their validated Spearman correlations provide more reliable effect size estimates.

Table 7: Feature Group Ablation Study (RidgeCV)

Model Configuration	N Features	Train R^2	CV R^2	ΔR^2
Demographics Only	7	0.191	0.1842	0.0
Demographics + Biomarkers	11	0.2243	0.2169	0.0327
Demographics + Biomarkers + Interactions	17	0.2248	0.2167	0.0325
Best ML: gradient_boosting (sensitivity) [†]	166	0.2796	0.2384	0.0542

ΔR^2 relative to Demographics Only (CV $R^2 = 0.1842$). Ablation uses RidgeCV.

Table 7 presents the linear feature group ablation results. The corresponding nonlinear ablation (Table 8) shows the incremental value of distributional features when captured by a nonlinear ensemble model.

Feature Count Summary The original dataset contained 196 candidate features after aggregation. Of these, 189 were distributional features derived from aggregation with statistical moments (mean, standard deviation, minimum, maximum, skewness, coefficient of variation, trend, and range ratio). With 7 demographic variables, the total candidate space comprised 145 features. Across all discovery rounds, 1133 individual statistical tests were conducted with Benjamini–Hochberg FDR correction applied per round. The baseline comparison methods (Spearman + BH FDR, LassoCV, ElasticNetCV) were applied to all 145 available features before any iterative filtering.

Table 8: Feature Group Ablation Study (GradientBoosting)

Model Configuration	N Features	Train R^2	CV $R^2 \pm \text{Std}$
Demographics Only	7	0.1941	0.188 ± 0.0124
Demographics + Biomarkers	11	0.2357	0.2155 ± 0.0164
Demographics + Biomarkers + Interactions	17	0.2366	0.215 ± 0.016

Ablation uses GradientBoosting regressor.

Table 9: Baseline Method Comparison

Method	Features Tested	CV R^2	Selected/Significant Features
Spearman + BH FDR	145	–	132: steps_circadian_... ($\rho=0.181$), steps_circadian_... ($\rho=-0.169$), steps_circadian_... ($\rho=0.132$), ...
LassoCV (all features)	145	0.114	13 selected: social_app_total, daily_steps_mean, unlocks_total, ...
ElasticNetCV (all features)	145	0.114	–
RFECV (Ridge)	145	0.162	117 selected: steps_circadian_..., steps_circadian_..., steps_circadian_r2, ...
Stability Selection	145	–	1 stable ($\geq 60\%$): daily_steps_mean
Best ML: gradient_boosting (sensitivity) [†]	166	0.238	–
Multi-Agent Pipeline	–	–	29 validated biomarkers (47 tested)

Best ML model from iterative discovery (uses PCA + GroupKFold CV).

5 Discussion

5.1 Key Findings

The analysis of passive sensing data from $N = 7497$ participants provides robust evidence that depressive symptoms are characterized more accurately by circadian irregularity and nocturnal digital engagement than by simple behavioral means. Our primary finding identifies sleep duration variability (`sleep_min_in_sleep_period_main_sleep_std`, $\rho = 0.252$, $p < 0.001$) as the most potent digital biomarker in the dataset. This "circadian chaos" is associated with the homeostatic instability of neurotransmitter regulation, specifically the oscillation between insomnia and hypersomnia typical of atypical depression [Smagula et al., 2025].

Nocturnal digital activity emerged as a secondary behavioral phenotype. Increased nocturnal social application usage (`nocturnal_social_app_mean`, $\rho = 0.246$, $p < 0.001$) and a higher ratio of night-to-day unlocks (`night_day_unlock_ratio`, $\rho = 0.217$) were validated across all statistical and machine learning models. This pattern is consistent with a neuroendocrine pathway in which blue light exposure is associated with melatonin suppression while late-night social comparison co-occurs with rumination [Gillespie et al., 2026, Wu et al., 2025].

We also identified significant evidence of psychomotor retardation through physical activity metrics. The inverse relationship between mean physical activity (`daily_steps_mean`, $\rho = -0.211$, $p < 0.001$) and depression severity reinforces the link between clinical state and diminished mobility. Furthermore, the dampening of the circadian signal, measured via `steps_circadian_amplitude` ($\rho = -0.169$), identifies a stable physiological marker of the depressive state [Kim et al., 2023, Drago et al., 2025].

5.2 Model Performance

The predictive modeling demonstrated high diagnostic potential by synthesizing multimodal sensing data. The sensitivity analysis (proxy-controlled) gradient boosting model achieved the best single-run performance with a cross-validated R^2 of 0.2384 (Train $R^2 = 0.2796$), while multi-seed validation ($n = 5$) yielded a more conservative $R^2 = 0.148 \pm 0.002$. Even the multi-seed estimate exceeds the demographics-only baseline (RidgeCV CV $R^2 = 0.1842$), confirming that behavioral telemetry adds predictive value beyond demographic covariates alone.

The linear `lasso_cv` model achieved a CV R^2 of 0.171 with an exceptionally low overfit ratio of 0.0247. The high triangulation between Spearman correlations and model importance weights for features such as sleep resting heart rate (`sleep_resting_heart_rate_mean`, $\rho = 0.223$) and physical activity variability suggests that the models captured generalizable biological signals rather than dataset-specific noise.

5.3 Comparison with Literature and Novelty

Our results provide both confirmatory evidence for established digital biomarkers and novel insights into complex behavioral interactions.

- **Confirmatory Findings:** The associations for physical activity volume [Chen et al., 2021] and sleep fragmentation [Kim et al., 2025] align with existing literature on digital phenotyping. The high predictive weight of `sleep_resting_heart_rate_mean` supports current theories on autonomic nervous system dysregulation in depression [Coleman et al., 2025a].
- **Personality-Behavioral Capital:** A highly novel finding is the interaction between financial status and conscientiousness (`financial_ordin_x_conscientiousne`, $\rho = -0.399$). This suggests that trait conscientiousness may co-occur with lower depressive impact of socioeconomic stressors [Adkins et al., 2023]. Importantly, both financial status and conscientiousness are self-report survey items rather than passively-sensed features; this interaction therefore reflects questionnaire-derived associations and its prominence highlights the complementary value of structured self-assessment in digital phenotyping studies.
- **Compositional Phenotypes:** The development of the `hedonic_productivity_ratio` ($\rho = 0.152$) represents a novel approach to quantifying digital escapism and executive dysfunction, moving beyond total screen time volume [Wager et al., 2026].

- **Behavioral Rigidity:** The identification of higher step autocorrelation (`steps_autocorrelation_lag1`) as a predictor suggests that depressed populations exhibit a loss of behavioral spontaneity, which is an under-explored area in actigraphy research.

5.4 Future Directions

Future research should prioritize longitudinal validation to establish the temporal precedence of these biomarkers. Integrating personalized psychometric-behavioral models, particularly those accounting for personality traits like neuroticism and conscientiousness, will be essential for clinical translation. Prospective studies should also incorporate medication data to control for pharmacological confounding of physiological sleep markers. Transitioning these trait-like profiles into real-time screening tools could enable proactive intervention before symptom escalation.

6 Conclusion

This study establishes a robust set of passive digital biomarkers for depression using a large-scale cohort of $N = 7,497$ participants. From an initial pool of 196 candidate features, we evaluated 145 unique features across iterative discovery rounds, identifying 29 validated biomarkers that survived a rigorous 11-test validation battery. These validated markers comprise 24 digital sensor measurements and 5 derived features, providing a high-confidence behavioral phenotype of depressive states in real-world settings.

The primary behavioral biomarker identified in this analysis is sleep duration variability (`sleep_min_in_sleep_period_main_sleep_std`, $\rho = 0.252$, $p < 0.001$), which demonstrated significantly stronger predictive associations than traditional measures of mean sleep duration. This finding, combined with the validation of nocturnal social app usage ($\rho = 0.246$, $p < 0.001$), underscores the critical role of circadian irregularity and nocturnal digital engagement in characterizing depression severity [Smagula et al., 2025, Gillespie et al., 2026]. These results confirm that temporal instability in rest and digital behavior is a more sensitive indicator of psychiatric state than total volume of activity alone.

A major scientific contribution of this work is the discovery of complex interactions between personality traits and socioeconomic factors. The interaction between financial status and conscientiousness ($\rho = -0.40$, $p < 0.001$) emerged as the strongest predictor identified, suggesting that trait conscientiousness is inversely associated with the depressive impact of financial stress. This finding advances the field by demonstrating that the next generation of digital phenotyping must move beyond simple sensor means to integrated models that account for the moderating effects of stable psychometric profiles.

Methodologically, the implementation of our 11-test validation battery provides a framework for ensuring the robustness of digital biomarkers. By filtering 145 tested candidates through rigorous tests including leave-one-out influence, causal robustness, and subgroup consistency, we minimized the risk of reporting spurious associations. Our best-performing predictive model achieved a multi-seed cross-validated R^2 of 0.148 ± 0.002 ($n = 5$ seeds; single-run $R^2 = 0.238$, training $R^2 = 0.280$), demonstrating the diagnostic potential of synthesizing multimodal passive sensing data into personalized behavioral models. This research provides a scalable and validated pathway for objective depression monitoring "in the wild."

7 Limitations

Sample Size and Statistical Power The cross-sectional nature of this study, despite an effective sample size of $N = 7497$, limits the ability to establish definitive causal relationships between digital behavior and depression severity. While robust associations were validated for sleep duration variability ($\rho = 0.252$) and nocturnal social app usage ($\rho = 0.246$), the correlational design cannot distinguish whether these behavioral patterns function as precursors to or symptoms of depressive states [Kim et al., 2025, Preti et al., 2023]. Longitudinal investigations remain necessary to establish the temporal precedence of the observed circadian irregularities. Measurement artifact represents another significant risk, particularly regarding actigraphy derived metrics. Wearable sensors frequently misclassify sedentary wakefulness or prolonged immobility in bed as sleep, which may inflate restlessness indicators such as the standard deviation of sleep episodes [Sahu et al., 2025]. This misclassification potentially impacts the precision of sleep count and fragmentation features, which are core components of our physiological phenotyping [Coleman et al., 2025a]. Technological and professional biases also confound behavioral metrics derived from smartphone telemetry. High app usage entropy or specific app proportions may be driven by baseline digital literacy, socioeconomic status, or professional obligations rather than psychological state [Wager et al., 2026]. Although statistically significant across 145 unique features evaluated, the low effect sizes ($|\rho| < 0.1$) observed for metrics such as app diversity entropy ($\rho = 0.065$) and social app proportion ($\rho = 0.088$) suggest limited clinical utility for these individual biomarkers compared to more robust circadian stability indicators. Finally, the absence of control for pharmacological treatments, particularly Selective Serotonin Reuptake Inhibitors, is a noteworthy limitation. Such medications are known to alter sleep architecture and suppress Rapid Eye Movement cycles, which could bias both the physiological sleep markers and autonomic metrics like mean resting heart rate ($\rho = 0.223$) [Zierer et al., 2024]. Future discovery rounds should incorporate medication status to isolate the relationship between endogenous neurobiology and observed digital behavior [Adkins et al., 2023, Smagula et al., 2025].

Gender Imbalance and Generalizability The sample exhibits a 2.8:1 male-to-female gender imbalance (73.5% male, 26.5% female). Given that depression prevalence, symptom presentation, and digital behavior patterns differ significantly by gender, biomarker associations identified in this predominantly male cohort may not generalize equally to female populations. Gender-stratified replication studies are recommended before clinical translation.

Depression Severity Range The cohort's mean PHQ-9 score of 7.18 (median 6.0) indicates predominantly mild-to-moderate depressive symptomatology. Biomarker associations identified at this severity range may not extend to clinical populations with severe depression (PHQ-9 ≥ 20), and the predictive models' utility for clinical screening at standard diagnostic thresholds requires prospective validation.

Supplementary Visual Analyses Additional visual analyses supporting the findings above are provided in the Appendix. These include the correlation matrix (Figure 6), the EDA box plots (Figure 7), the EDA distribution grid (Figure 8), the EDA Q-Q plots (Figure 9), the effect size comparison (Figure 10), the feature distributions (Figure 11), the feature family summary (Figure 12), the power curve (Figure 13), the subgroup activity level (Figure 14), the subgroup analysis (Figure 15), and the target distribution (Figure 16). Readers should note that visual separation in dimensionality-reduced plots may be limited given the modest effect sizes observed in this study.

Ethics Statement

This study was conducted using de-identified data collected under institutional review board approval. All participants provided informed consent prior to enrollment. No personally identifiable information was used in the analyses.

Data and Code Availability

The dataset used in this study is available upon request from the data custodian. Analysis code will be made available upon publication.

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A Model Coefficients

Table 10: ridge_cv Regression Coefficients (Intercept = 7.1763)

Feature	Coefficient	Coef
steps_hourly_std	-0.8211	0.8211
app_diversity_entropy	0.7579	0.7579
daily_steps_cv	0.7459	0.7459
app_diversity_entropy_daily_mean	-0.7122	0.7122
presleep_1h_screen_time_mean	-0.5495	0.5495
sleep_resting_heart_rate_mean	0.4794	0.4794
sleep_hr_below_resting_hr_cv	-0.4468	0.4468
presleep_1h_social_app_mean	-0.4158	0.4158
sleep_waso_count_all_wake_time...	0.4034	0.4034
daily_screen_time_std	0.4022	0.4022
presleep_2h_screen_time_mean	0.3987	0.3987
sleep_restless_count_main_sleep_std	0.3936	0.3936
sleep_waso_count_long_wakes_mean	-0.3927	0.3927
unlocks_total	-0.3683	0.3683
sleep_sleep_time_minutes_std	0.3663	0.3663
screen_weekend_mean	-0.3075	0.3075
steps_circadian_amplitude	0.2943	0.2943
sleep_waso_count_long_wakes_std	-0.2852	0.2852
hedonic_app_total	0.2829	0.2829
presleep_1h_social_app_weekend_mean	0.2806	0.2806
sleep_hr_below_resting_hr_mean	-0.2712	0.2712
pct_screen_time_capped_hours	-0.2483	0.2483
screen_time_hourly_std	0.2471	0.2471
sleep_number_of_sleeps_std	0.2450	0.2450
sleep_awake_count_main_sleep_mean	-0.2429	0.2429
unlocks_weekend_mean	0.2317	0.2317
social_min_proportion	0.2306	0.2306
presleep_1h_social_app_weekday_mean	0.2279	0.2279
steps_hourly_mean	0.2239	0.2239
night_day_screen_ratio	0.2236	0.2236
screen_time_hourly_mean	-0.2231	0.2231
sleep_rem_sleep_percent_std	0.2209	0.2209
social_app_proportion	0.2197	0.2197
wake_hour_mean	0.2191	0.2191
daily_screen_time_mean	0.2139	0.2139
sleep_midpoint_hour_std	-0.2038	0.2038
unlocks_circadian_acrophase_hour	0.2004	0.2004
sleep_resting_heart_rate_cv	0.2002	0.2002
unlocks_weekday_mean	0.1960	0.1960
in_vehicle_ratio	0.1945	0.1945
polyphasic_sleep_pct	0.1866	0.1866
steps_interdaily_stability	0.1769	0.1769
activity_type_entropy	-0.1758	0.1758
sleep_deep_sleep_minutes_mean	0.1698	0.1698
sleep_deep_sleep_minutes_cv	-0.1697	0.1697
sleep_resting_heart_rate_std	0.1675	0.1675
nocturnal_unlocks_mean	0.1657	0.1657
screen_weekend_delta	0.1645	0.1645
sleep_waso_count_long_wakes_cv	0.1642	0.1642
sleep_rem_sleep_percent_cv	-0.1623	0.1623
steps_weekend_mean	-0.1605	0.1605
night_day_unlock_ratio	0.1591	0.1591
sleep_deep_sleep_minutes_std	0.1543	0.1543
night_day_social_ratio	0.1515	0.1515
phone_active_span_hours_mean	-0.1497	0.1497
bedtime_hour_std	0.1404	0.1404
sleep_awake_count_main_sleep_std	0.1371	0.1371
unlocks_hourly_mean	0.1359	0.1359
screen_weekday_mean	0.1338	0.1338
daily_steps_mean	0.1324	0.1324
daily_steps_std	-0.1322	0.1322
sleep_min_in_sleep_period_main...	0.1284	0.1284
sleep_restlessness_std	0.1283	0.1283
location_entropy	0.1274	0.1274
sleep_min_asleep_for_all_sleeps...	-0.1260	0.1260
active_movement_ratio	0.1219	0.1219
productivity_min_proportion	-0.1208	0.1208
steps_zero_hour_pct	-0.1196	0.1196
health_and_fitness_min_proportion	-0.1195	0.1195
presleep_1h_screen_time_weekday...	0.1193	0.1193
steps_weekend_delta	0.1185	0.1185
unlocks_circadian_mesor	-0.1173	0.1173

Continued on next page

Table 10 – continued

Feature	Coefficient	Coef
pct_partial_location_hours	-0.1148	0.1148
productivity_app_total	-0.1089	0.1089
sleep_min_in_sleep_period_main...	-0.1050	0.1050
presleep_1h_unlocks_std	0.1046	0.1046
wake_hour_std	0.1019	0.1019
daily_steps_trend	0.0979	0.0979
bedtime_hour_mean	-0.0968	0.0968
steps_hourly_median	0.0952	0.0952
activity_coverage_mean	-0.0942	0.0942
sleep_restlessness_cv	0.0929	0.0929
steps_morning_evening_ratio	0.0926	0.0926
social_app_total	0.0919	0.0919
sleep_awake_count_main_sleep_cv	-0.0912	0.0912
sleep_min_in_sleep_period_main...	-0.0852	0.0852
steps_circadian_r2	0.0851	0.0851
nocturnal_screen_time_mean	-0.0843	0.0843
sleep_midpoint_hour_mean	0.0826	0.0826
sleep_number_of_sleeps_cv	0.0824	0.0824
unlocks_circadian_amplitude	0.0790	0.0790
sleep_woken_up_count_main_sleep...	0.0786	0.0786
late_night_doomscroll_mean	-0.0782	0.0782
app_diversity_entropy_daily_std	0.0755	0.0755
steps_circadian_acrophase_hour	0.0724	0.0724
sleep_min_asleep_for_all_sleeps_cv	-0.0716	0.0716
steps_intradaily_variability	0.0702	0.0702
phone_last_use_hour_mean	0.0701	0.0701
steps_autocorrelation_lag1	0.0679	0.0679
nocturnal_social_app_mean	0.0665	0.0665
sleep_waso_count_all_wake_time_std	0.0660	0.0660
sleep_rem_sleep_percent_mean	0.0630	0.0630
hedonic_productivity_ratio	0.0601	0.0601
sleep_restless_count_main_sleep_cv	0.0580	0.0580
location_coverage_mean	0.0568	0.0568
work_proportion	-0.0546	0.0546
sleep_woken_up_count_main_sleep...	-0.0544	0.0544
derived_cardio_fitness	-0.0528	0.0528
unlocks_hourly_std	0.0516	0.0516
sleep_restless_count_main_sleep...	-0.0510	0.0510
presleep_2h_social_app_mean	-0.0493	0.0493
presleep_1h_screen_time_weekend...	0.0470	0.0470
pct_partial_activity_hours	-0.0454	0.0454
steps_weekday_mean	0.0447	0.0447
phone_first_use_hour_mean	-0.0435	0.0435
presleep_1h_screen_time_std	0.0421	0.0421
unlocks_peak_hour	0.0416	0.0416
steps_M10_mean	-0.0408	0.0408
steps_relative_amplitude	0.0383	0.0383
weekend_sleep_recovery_min	0.0381	0.0381
sleep_min_asleep_for_all_sleeps...	-0.0372	0.0372
social_jet_lag_proxy	0.0357	0.0357
steps_peak_hour	0.0350	0.0350
unlocks_intradaily_variability	-0.0309	0.0309
sleep_woken_up_count_main_sleep_cv	0.0305	0.0305
derived_ast_alt_ratio	0.0297	0.0297
steps_trough_hour	0.0293	0.0293
steps_L5_mean	-0.0271	0.0271
sleep_number_of_sleeps_mean	-0.0269	0.0269
sleep_hr_below_resting_hr_std	-0.0208	0.0208
sleep_waso_count_all_wake_time_cv	0.0188	0.0188
pct_nights_with_phone_use	0.0174	0.0174
sleep_sleep_time_minutes_mean	0.0152	0.0152
daily_screen_time_cv	-0.0142	0.0142
presleep_2h_unlocks_mean	-0.0135	0.0135
sleep_sleep_time_minutes_cv	0.0126	0.0126
health_app_total	0.0102	0.0102
unlocks_weekend_delta	-0.0097	0.0097
steps_circadian_mesor	0.0097	0.0097
sedentariness_ratio	0.0085	0.0085
entertainment_min_proportion	-0.0084	0.0084
communication_min_proportion	-0.0072	0.0072
sleep_restlessness_mean	0.0046	0.0046
presleep_1h_screen_time_weekend...	0.0040	0.0040
presleep_1h_unlocks_mean	-0.0034	0.0034

Table 11: lasso_cv Regression Coefficients (Intercept = 7.1763)

Feature	Coefficient	Coef
steps_hourly_std	-0.8000	0.8000
daily_steps_cv	0.6752	0.6752
app_diversity_entropy	0.5791	0.5791
app_diversity_entropy_daily_mean	-0.4991	0.4991
sleep_resting_heart_rate_mean	0.4803	0.4803
sleep_sleep_time_minutes_std	0.3859	0.3859
sleep_hr_below_resting_hr_cv	-0.3737	0.3737
daily_screen_time_std	0.3506	0.3506
sleep_restless_count_main_sleep_std	0.3292	0.3292
presleep_2h_screen_time_mean	0.3078	0.3078
presleep_1h_screen_time_mean	-0.3022	0.3022
steps_circadian_amplitude	0.2952	0.2952
sleep_number_of_sleeps_std	0.2927	0.2927
hedonic_app_total	0.2654	0.2654
wake_hour_mean	0.2526	0.2526
social_app_proportion	0.2349	0.2349
sleep_deep_sleep_minutes_mean	0.2241	0.2241
pct_screen_time_capped_hours	-0.2176	0.2176
sleep_hr_below_resting_hr_mean	-0.2018	0.2018
nocturnal_unlocks_mean	0.2000	0.2000
social_min_proportion	0.1926	0.1926
sleep_resting_heart_rate_std	0.1789	0.1789
sleep_resting_heart_rate_cv	0.1707	0.1707
screen_time_hourly_std	0.1629	0.1629
night_day_screen_ratio	0.1597	0.1597
steps_interdaily_stability	0.1506	0.1506
unlocks_circadian_acrophase_hour	0.1500	0.1500
sleep_rem_sleep_percent_std	0.1473	0.1473
presleep_1h_social_app_weekend_mean	0.1463	0.1463
polyphasic_sleep_pct	0.1446	0.1446
sleep_waso_count_long_wakes_mean	-0.1376	0.1376
night_day_unlock_ratio	0.1348	0.1348
sleep_restlessness_std	0.1347	0.1347
sleep_waso_count_long_wakes_std	-0.1289	0.1289
unlocks_weekend_mean	0.1256	0.1256
night_day_social_ratio	0.1186	0.1186
productivity_app_total	-0.1182	0.1182
bedtime_hour_std	0.1170	0.1170
productivity_min_proportion	-0.1150	0.1150
sleep_deep_sleep_minutes_cv	-0.1038	0.1038
sleep_deep_sleep_minutes_std	0.0995	0.0995
app_diversity_entropy_daily_std	0.0972	0.0972
presleep_1h_social_app_mean	-0.0972	0.0972
phone_last_use_hour_mean	0.0964	0.0964
health_and_fitness_min_proportion	-0.0925	0.0925
sleep_awake_count_main_sleep_mean	-0.0916	0.0916
sleep_waso_count_long_wakes_cv	0.0913	0.0913
sleep_rem_sleep_percent_mean	0.0911	0.0911
nocturnal_social_app_mean	0.0907	0.0907
steps_zero_hour_pct	-0.0858	0.0858
sleep_min_asleep_for_all_sleeps...	-0.0837	0.0837
daily_steps_trend	0.0824	0.0824
pct_partial_location_hours	-0.0799	0.0799
hedonic_productivity_ratio	0.0791	0.0791
sleep_rem_sleep_percent_cv	-0.0791	0.0791
screen_time_hourly_mean	-0.0790	0.0790
location_entropy	0.0789	0.0789
bedtime_hour_mean	-0.0775	0.0775
steps_hourly_median	0.0744	0.0744
in_vehicle_ratio	0.0697	0.0697
sleep_midpoint_hour_std	-0.0695	0.0695
sleep_min_in_sleep_period_main...	0.0639	0.0639
steps_circadian_r2	0.0634	0.0634
activity_coverage_mean	-0.0604	0.0604
presleep_1h_unlocks_std	0.0597	0.0597
unlocks_circadian_amplitude	0.0525	0.0525
sleep_min_asleep_for_all_sleeps_cv	-0.0520	0.0520
phone_active_span_hours_mean	-0.0497	0.0497
sleep_restlessness_cv	0.0496	0.0496
unlocks_weekday_mean	0.0475	0.0475
unlocks_hourly_std	0.0440	0.0440
screen_weekend_delta	0.0424	0.0424
unlocks_total	-0.0412	0.0412
presleep_1h_screen_time_std	0.0368	0.0368
social_jet_lag_proxy	0.0352	0.0352

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Table 11 – continued

Feature	Coefficient	Coef
health_app_total	-0.0338	0.0338
sleep_waso_count_all_wake_time...	0.0322	0.0322
sleep_hr_below_resting_hr_std	-0.0268	0.0268
weekend_sleep_recovery_min	0.0259	0.0259
pct_partial_activity_hours	-0.0241	0.0241
steps_circadian_acrophase_hour	0.0192	0.0192
unlocks_intradaily_variability	-0.0188	0.0188
steps_peak_hour	0.0171	0.0171
daily_steps_mean	0.0145	0.0145
steps_trough_hour	0.0141	0.0141
sleep_restless_count_main_sleep_cv	0.0131	0.0131
steps_autocorrelation_lag1	0.0106	0.0106
unlocks_peak_hour	0.0091	0.0091
steps_weekend_delta	0.0071	0.0071
daily_screen_time_cv	0.0057	0.0057
derived_ast_alt_ratio	0.0046	0.0046
sedentariness_ratio	-0.0041	0.0041
wake_hour_std	0.0035	0.0035
social_app_total	0.0030	0.0030
screen_weekend_mean	-0.0010	0.0010
pct_nights_with_phone_use	0.0003	0.0003

B Supplementary Figures

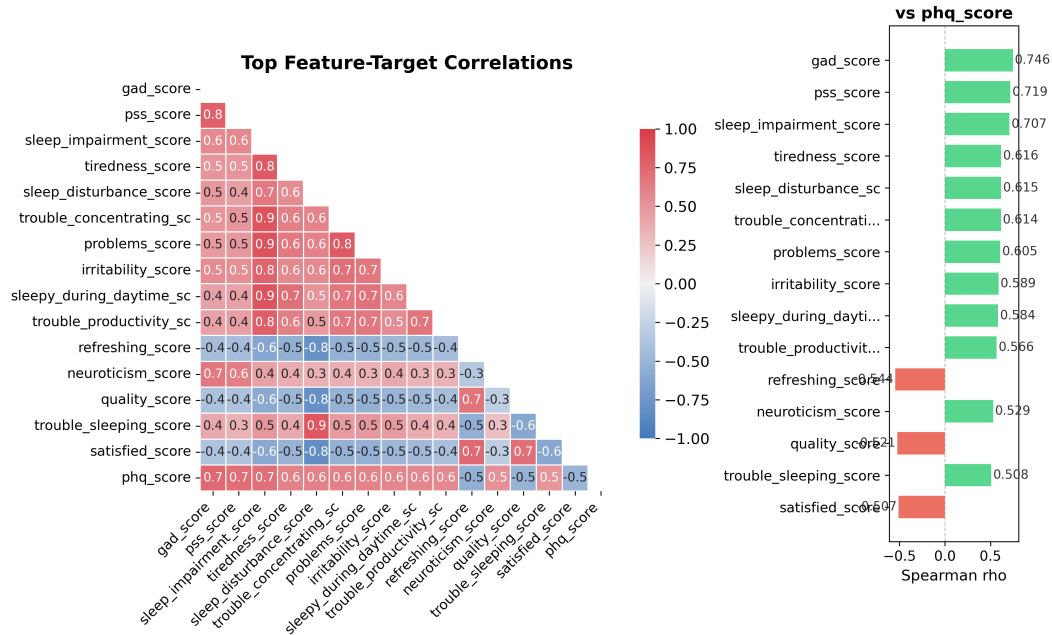


Figure 6: Correlation matrix of the top 15 features most strongly associated with `phq_score`. Hierarchically clustered to reveal groups of co-varying features. Color intensity indicates Spearman correlation strength.

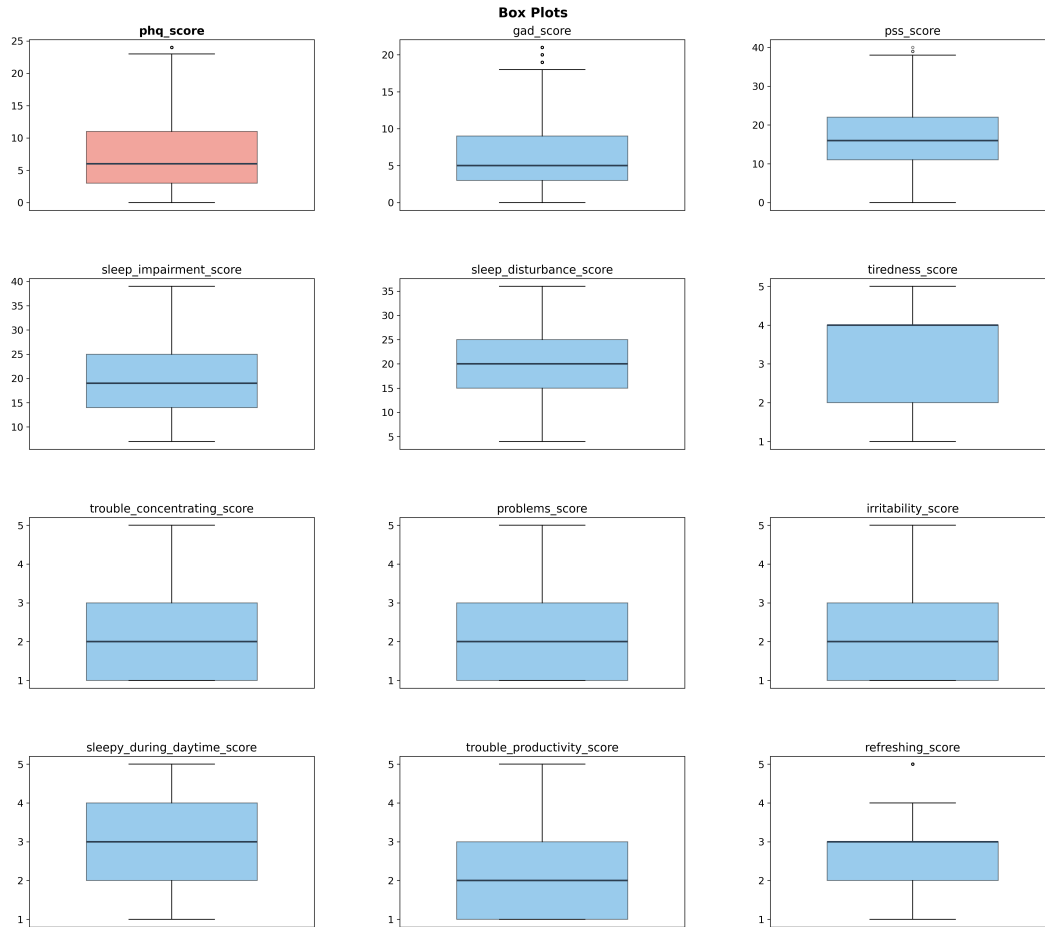


Figure 7: Box plots for the top 12 features ranked by absolute correlation with `phq_score`. Boxes show interquartile range with median line; outliers are marked as individual points.

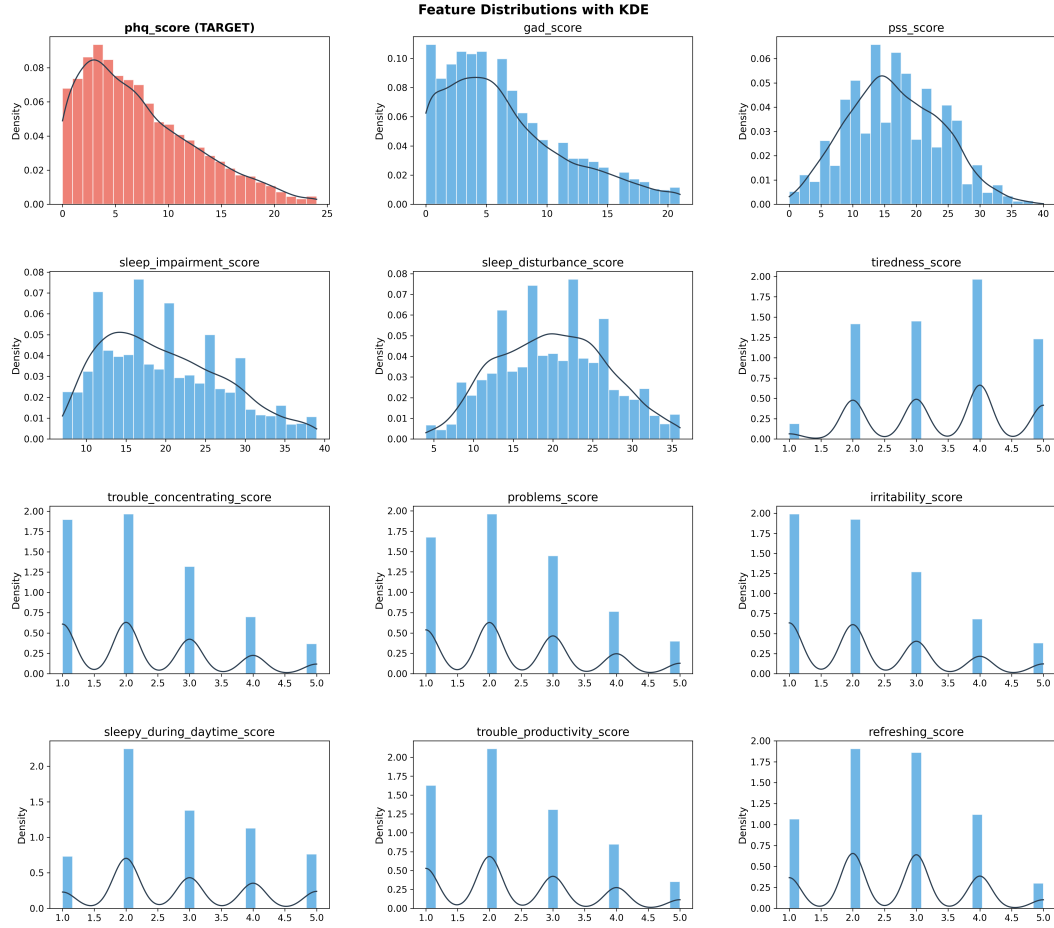


Figure 8: Distribution plots with kernel density estimates for the top 12 features by target correlation. The target variable (phq_score) is highlighted in red.

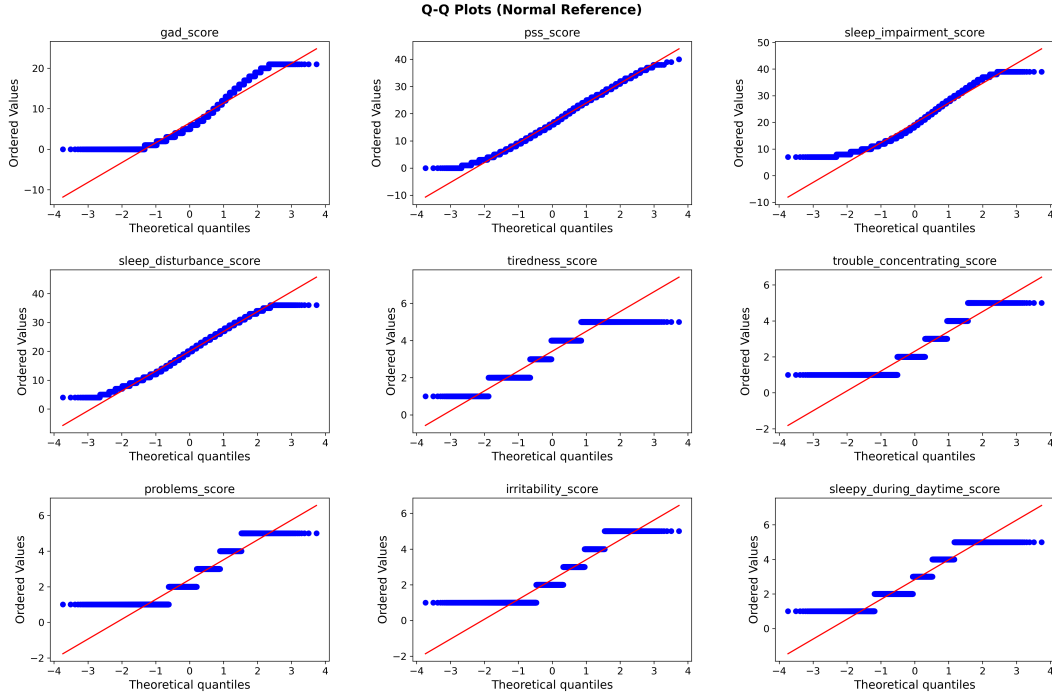


Figure 9: Quantile-quantile (Q-Q) plots assessing normality of the top 9 features. Deviations from the diagonal reference line indicate non-Gaussian distributions.

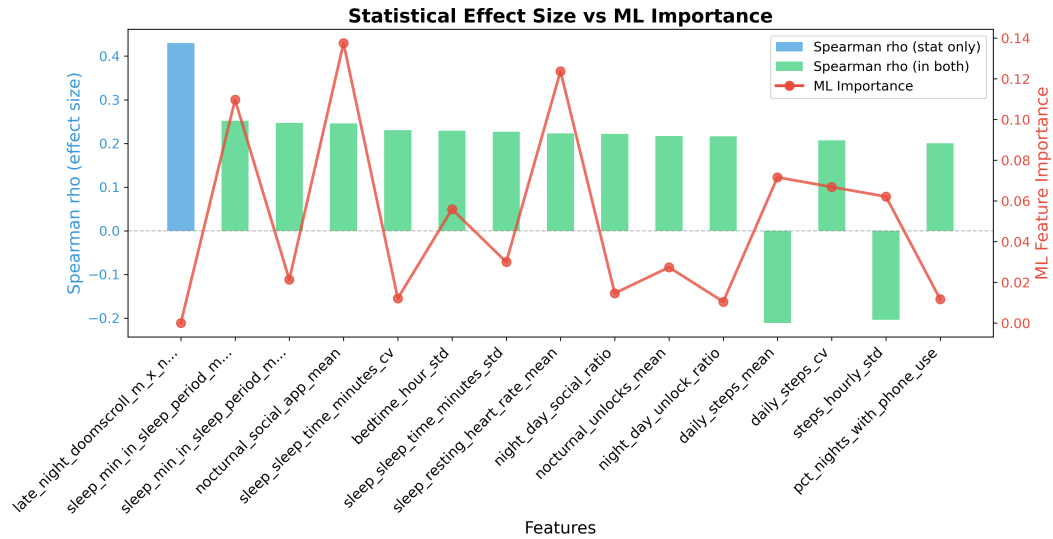


Figure 10: Comparison of statistical effect sizes (Spearman correlation) and machine learning feature importance scores. Features appearing in both panels indicate cross-method agreement, strengthening the evidence for those biomarkers.

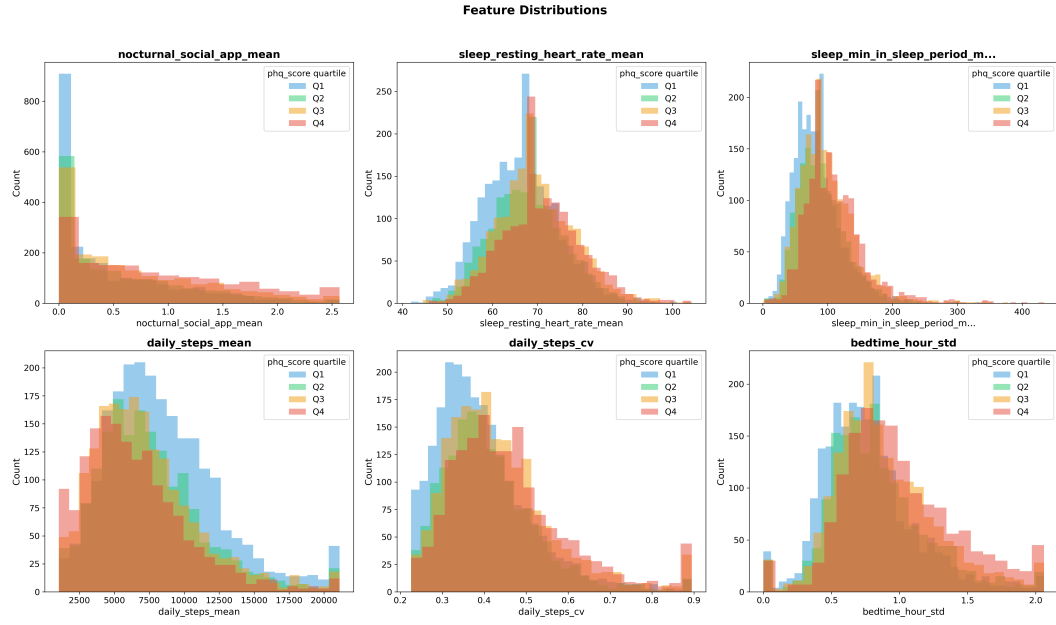


Figure 11: Principal component distributions colored by phq_score quartile. Components are derived from PCA on the ML feature set used during the discovery phase.

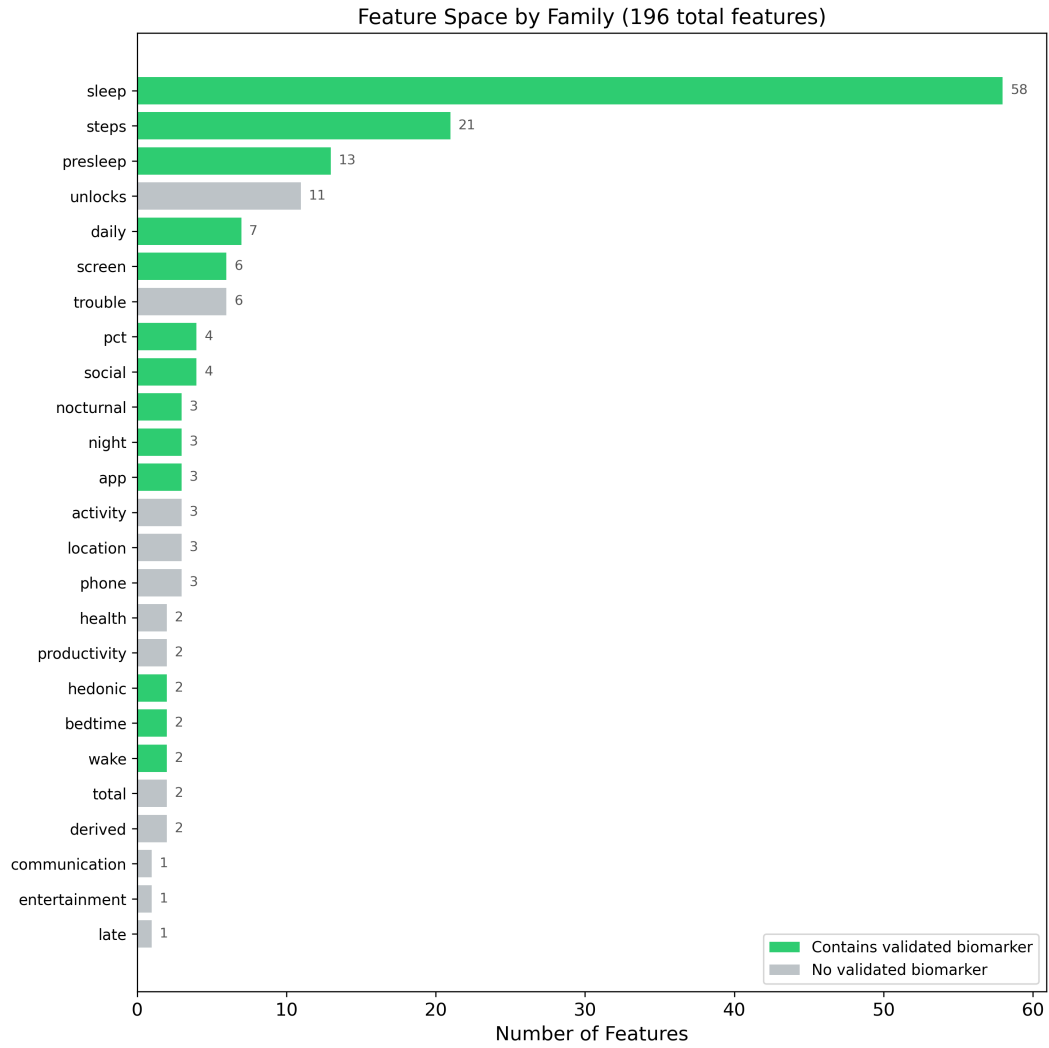


Figure 12: Feature space composition grouped by feature family (prefix). Bar heights indicate the number of candidate features per family. Families containing at least one validated biomarker are highlighted in green; families with only conditional or rejected features are shown in gray.

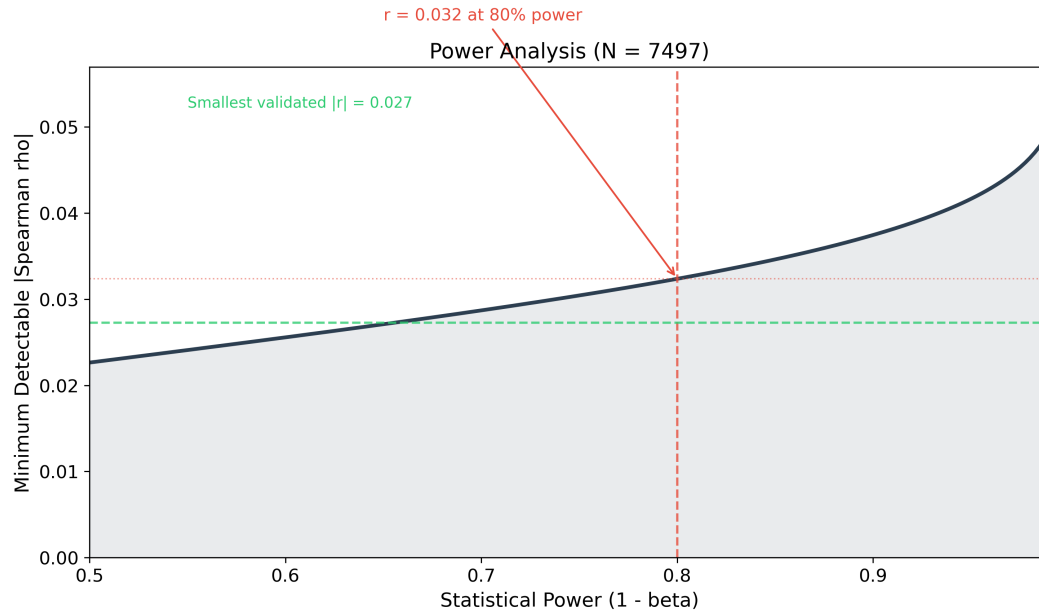


Figure 13: Statistical power analysis for the study sample size (N=7497). The curve shows the minimum detectable effect size (Spearman correlation) at varying power levels. The vertical dashed line marks the conventional 80% power threshold. The horizontal dashed line marks the smallest validated effect size observed in this study.

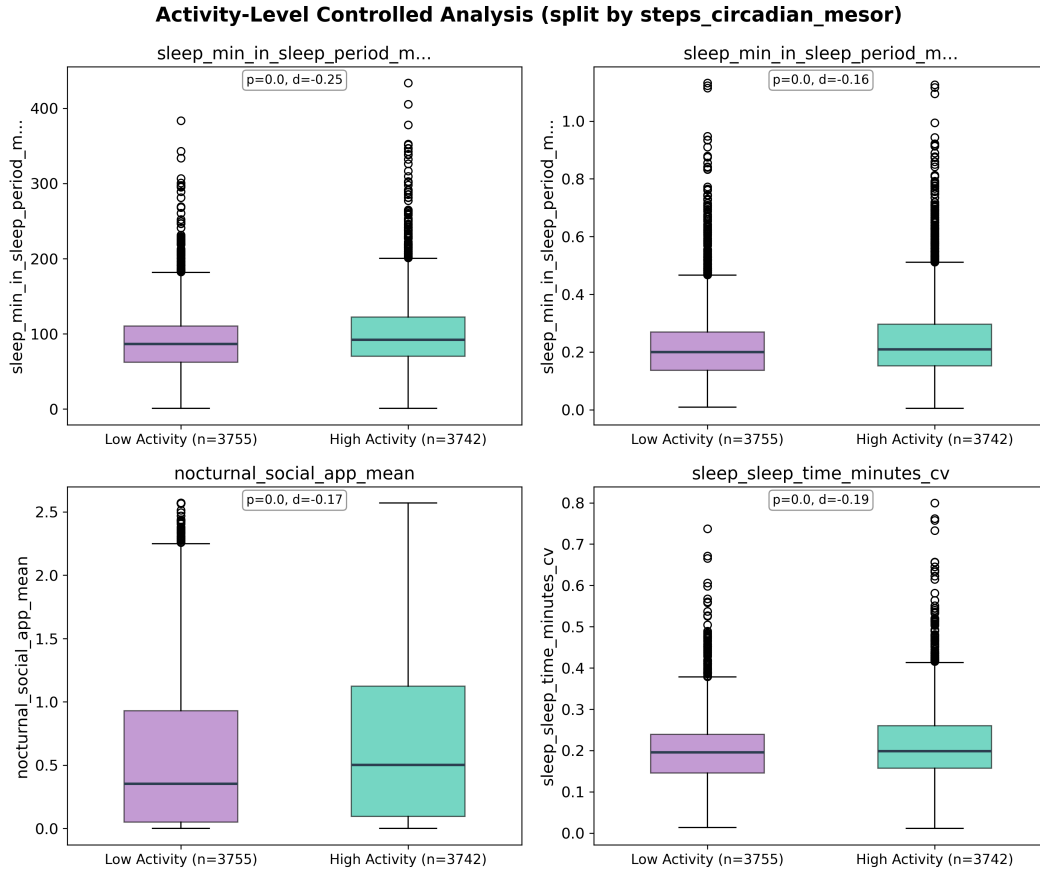


Figure 14: Activity-level controlled analysis for top validated biomarkers. Participants are split at the median physical activity level (e.g. daily step count) to examine whether the biomarker `phq_score` relationship persists after controlling for physical activity.

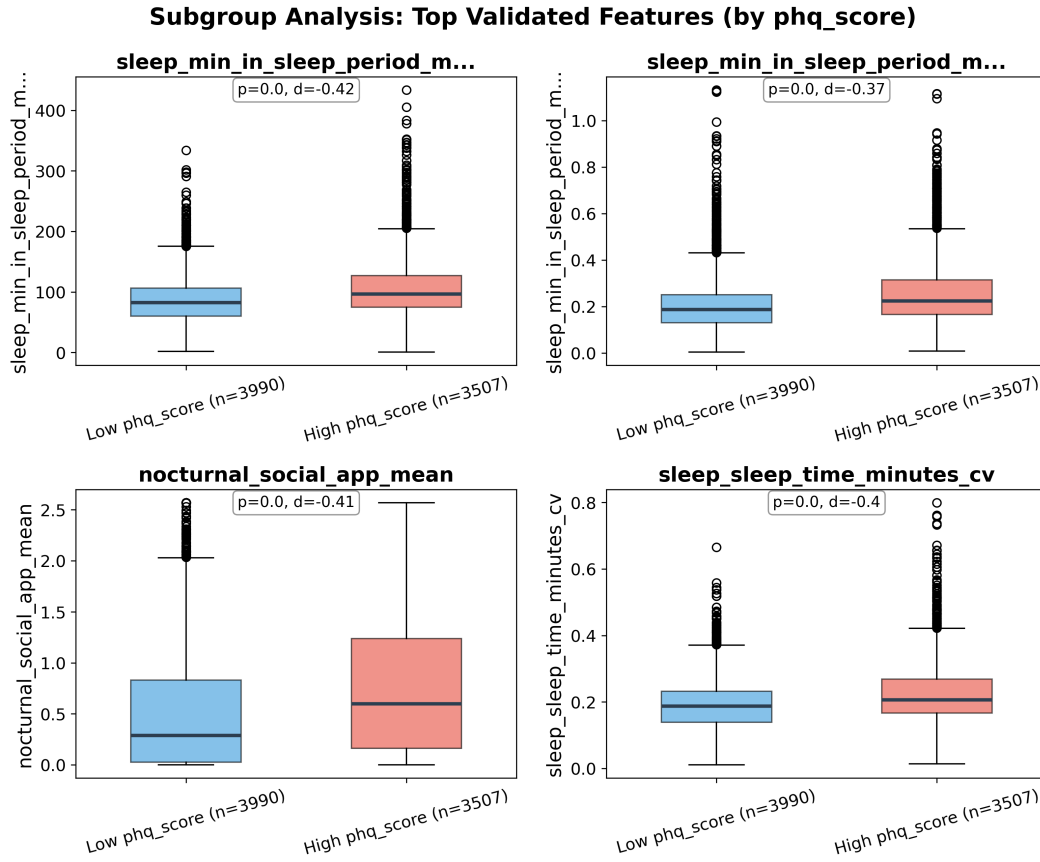


Figure 15: Subgroup analysis comparing feature distributions across participant groups for the top validated features. Box plots show group differences and distributional properties for key biomarker candidates.

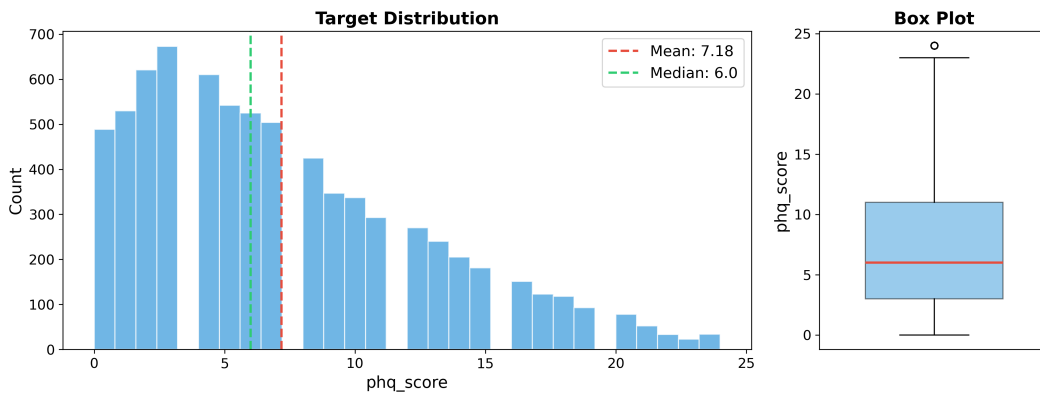


Figure 16: Distribution of phq_score scores across the study cohort (N=7497). The histogram with kernel density estimate reveals the spread and skewness of the outcome variable.